Out-of-Sample Evaluation of Textual News Sentiment for Predicting the Equity Premium and Consumer Sentiment

SentCast: GDELT Sentiment on S&P 500

Vignesh Nagarajan

Texas A&M University

vigneshn26@tamu.edu

Abstract

Textual sentiment extracted from news is widely proposed as a leading indicator for financial markets and the real economy, yet most proposed predictors of the equity premium fail out of sample. We put news sentiment through an out-of-sample test in which the sample, the models, the tests and the decision rule were fixed before any experiment ran, and every planned experiment is reported. We ask whether a monthly sentiment index improves one-month-ahead forecasts of the equity premium (the S&P 500 excess return), its direction, and the change in the University of Michigan Index of Consumer Sentiment. The index combines four standard sentiment scorers, from a finance word list to FinBERT, applied to 125,105 economy and market headlines from the GDELT database for 2015 to 2026. Automated checks ensure that every headline predates its forecast date. Over 90 test months the index does not significantly improve on the benchmark for any of the three targets. For the equity premium the out-of-sample R-squared against the historical mean is 1.03% (Clark-West $p = 0.149$): positive, but not distinguishable from zero. A power simulation, added after the results were known, shows that the test reaches 80% power only for a true R-squared of at least 8.0%, so the result speaks against a large effect and leaves a small one open. Indices built from two large language models, and a published news sentiment index over a longer sample, do not significantly improve on the benchmarks either; at a three-month horizon the evidence for the equity premium is mixed. The contribution is a bounded null and a leakage-checked baseline for text-based forecasting that reruns from free data with one command.

Keywords: sentiment analysis; financial markets; stock return forecasting; consumer sentiment; out-of-sample evaluation; large language models

1 Introduction

News text is produced every day and is available at once, while most economic statistics arrive weeks after the period they describe. Text is now widely used as data in economics and finance [1], and its sentiment, the positive or negative tone of the words, is a natural candidate for an early signal about markets and the economy. Pessimism in a daily newspaper column predicts lower market prices followed by a reversal [2], and an index of newspaper sentiment predicts movements in consumer sentiment [3]. A recent proposal applies the idea to social media: it forecasts United States economic

behaviour from the sentiment of posts and reports forecasts of the S&P 500 for four months of 2024 [4]. That study lists an overlap between training and input data among its limitations and names a longer sample and a comparison of models as future work.

A longer sample makes the task hard. Most variables proposed as predictors of the aggregate stock return fail to beat the historical mean return in forecasts of months that were not used for fitting [5], and most predictors proposed since then lose significance or perform poorly in such forecasts [6]. With sign restrictions on the forecasts many predictors do beat the historical mean, and a small gain can matter to an investor [7]. A sentiment index built for prediction has forecast aggregate returns in and out of sample [8]. Whether sentiment from free news text adds to simple forecasts is therefore an empirical question, and the answer depends on how the test is run: on the length of the sample, on whether accuracy is measured on months the model has not seen, and on whether the rule for reading the results is set before they are known.

This paper runs that test for one sentiment index and three monthly targets. We ask four questions. (RQ1) Does a monthly sentiment index built from free news headlines improve one-month-ahead forecasts of the S&P 500, for its excess return and for its direction? (RQ2) Does the same index improve one-month-ahead forecasts of the change in the University of Michigan Index of Consumer Sentiment? (RQ3) How do the constructed indices compare with published indices as predictors, and do they add anything to them? (RQ4) Is the index a sound measurement: do scorers agree, is it reliable given how the news is sampled, and does it track published measures?

Four features define the test. The questions, the sample, the models, the statistical tests and a rule that maps the results to an outcome of positive, mixed or null were written down before any experiment was run; later changes are dated and listed in Section 4. All inputs are free and can be downloaded again, with one exception, a secondary comparison that uses two commercial language models as scorers. Every forecast is made month by month from data dated before the forecast date, and automated checks stop the computation if a headline dated on or after a forecast date enters that forecast. All twelve experiments of the plan are reported, whatever their result.

The paper makes four contributions. The first is a test with a rule fixed in advance and every outcome reported, including the null. The second is a way to build a long headline corpus from free data, by rebuilding headlines from the web addresses of articles, together with checks that the rebuilt headlines track page titles and that an index from a small sample of each day's news is reliable. The third is a comparison of word lists, a trained classifier, FinBERT and two large language models as scorers on the same forecasting task, next to published indices evaluated over 439 months. The fourth is a pipeline that reruns the study from raw downloads with one command.[1]

Under the rule the outcome is null for all three targets. Over 90 test months, from April 2019 to September 2026, adding the index to a forecast based on the historical mean lowers the sum of squared forecast errors for the S&P 500 excess return by 1.03%, with a 90% interval from −2.44% to 4.80% that includes zero. For the direction the models with the index predict a rise in every month, and for consumer sentiment the squared errors are larger with the index than without it. A simulation that was not part of the plan puts the null in proportion. On a sample of this size the test finds a true R-squared of 1% in 14.3% of simulated samples and reaches a power of 80% from 8.0%. The null therefore

[1]The code repository, with the result files behind every table and figure, is available at `https://github.com/vignesh-nagarajan-vn/SentCast`.

speaks against a large effect and leaves a small one open.

Section 2 places the study in the literature, Sections 3 and 4 describe the data and the method, Section 5 reports the results, Section 6 the robustness checks, the power of the test and the limitations, and Section 7 concludes.

2 Related work

Several studies of sentiment and the stock market work at the daily frequency. Tetlock [2] finds that pessimism in a daily newspaper column predicts lower market prices followed by a reversal. García [9] counts positive and negative words in New York Times columns from 1905 to 2005 and finds that the resulting sentiment predicts daily stock returns, mostly in recessions. Bollen et al. [10] report that some dimensions of Twitter mood improve forecasts of the daily direction of the Dow Jones index on a sample from 2008. Ke et al. [11] propose a supervised method that extracts the sentiment relevant for returns from news articles.

At the monthly frequency the reference point is the literature on predicting the aggregate stock return. Welch and Goyal [5] show that most proposed predictors fail out of sample against the historical mean, and Goyal et al. [6] reach a similar verdict for 29 newer predictors with data to 2021. Campbell and Thompson [7] show that sign restrictions let many predictors beat the historical mean and that a small out-of-sample R-squared can matter to investors. Huang et al. [8] build an investor sentiment index designed for prediction and find that it forecasts aggregate stock returns in and out of sample. Our evaluation follows this literature: we use the out-of-sample R-squared of Campbell and Thompson [7] and the test for nested models of Clark and West [12].

Text is also used to measure and forecast the economy; Gentzkow et al. [1] survey the methods. Shapiro et al. [3] build a sentiment index from newspaper articles for 1980 to 2015 and find that news sentiment predicts movements in consumer sentiment. Baker et al. [13] build an index of economic policy uncertainty from newspaper coverage, whose innovations foreshadow declines in investment, output and employment. Kalamara et al. [14] find that newspaper text improves forecasts of GDP, inflation and unemployment in the United Kingdom, most in periods of stress. The first two indices are published and updated, and we use them as benchmarks.

The scorers we use come from four lines of work. Loughran and McDonald [15] show that general word lists misclassify financial text and provide lists made for it. Hutto and Gilbert [16] provide VADER, a rule-based general sentiment lexicon validated on social media text. Malo et al. [17] provide the Financial PhraseBank, a set of financial sentences annotated by humans, and Araci [18] shows that a BERT language model further trained on financial text improves the classification of financial sentiment.

Large language models are a recent addition. Lopez-Lira and Tang [19] find that scores of news headlines from such a model predict stock returns after the model's knowledge cutoff. Glasserman and Lin [20] show that backtests of this kind are biased when the model's training window overlaps the test period. We therefore also report the language-model results for the months after a model's training cutoff.

Our text comes from GDELT, an open database built from world news [21]. The study differs from the daily studies above in its monthly frequency, and from the monthly return literature in its predictor: an index built only from free text that anyone can download again, with no vendor news feed. It takes

up the two items that the proposal in [4] leaves for future work, a longer sample and a comparison of models against baselines.

3 Data

3.1 Targets

All forecasts are dated on the last trading day of a calendar month, which we call the forecast date. The first target is the S&P 500 excess return of the following month: the log change of the index between two month-end closes, minus the return on a three-month Treasury bill over the same month, in percent,

$$R_m = 100 \left[\ln \frac{P_m}{P_{m-1}} - \ln \left(1 + \frac{b_{m-1}}{1200} \right) \right], \tag{1}$$

where P_m is the close on the last trading day of month m and b_{m-1} the bill rate of the previous month in percent per year. The excess return, also called the equity premium, is the standard target of the return prediction literature. The bill rate is known at the start of the month, and the index is a price index without dividends. The second target is the direction of the S&P 500: one if the month-end close is above the previous one and zero otherwise. The index rose in 63.3% of the test months. The third target is the monthly change in the University of Michigan Index of Consumer Sentiment, a household survey whose final value for a month is released at the end of that month or in the first days of the next. Month-end closes come from FRED, the data service of the Federal Reserve Bank of St. Louis, from November 2016 and from the data file of Welch and Goyal [5] for earlier months. The bill rate and the trading calendar also come from FRED.

The text sample gives 138 forecasts. The first is made at the end of March 2015 and the last at the end of August 2026. The first 48 serve only for the initial estimate, which leaves 90 test months, from April 2019 to September 2026.

3.2 Headline corpus

GDELT is an open database built from world news [21]. Its Global Knowledge Graph publishes one file every 15 minutes, with one row per article that holds the web address, the locations detected in the text and a tone score. The study is designed to run on one laptop, so we draw two of the 96 files of each day, from February 2015 to September 2026, with a random generator seeded by the date; days are counted in Coordinated Universal Time. A missing file is replaced by another file of the same day. The draw requested 8,484 files and obtained 8,450 (60.7 GB). Of these, 134 are replacements, and 34 draws on 17 days in June and July 2025 found no readable file.

Page titles are present in the files only for the later years: October 2019 is the first index month in which at least 95% of our headlines have one, and 53.4% of all headlines have one. To keep one definition for the whole sample we rebuild a headline from the words in the web address of each article. We keep an article if GDELT tags a United States location, the rebuilt headline has at least four words, it contains at least one of 54 fixed economy and market terms (inflation, recession, unemployment, Federal Reserve and Wall Street are examples), and its source is not one of 15 press-release sites. Exact duplicates are removed. Each headline carries the time of the file in which GDELT recorded the article, which is at or after its publication.

Table 1: The headline corpus: files drawn from GDELT, the filter funnel and the coverage of the index months.

Item	Value
Calendar days in the draw	4,242
Files requested	8,484
Files downloaded	8,450
Files replaced by another slot of the same day	134
Draws left without a file	34
Days with one file / with no file	0 / 17
Download volume (GB)	60.7
Article rows read	15,462,640
Web articles	15,451,737
With a US location	6,604,345
With a URL headline of 4+ words	5,641,583
With an economy or market term	164,612
Not from a press-release domain	160,636
After URL de-duplication	160,635
After same-day headline de-duplication	125,105
Index months	138
Headlines per index month: min / median / max	498 / 870.5 / 2,015
Valid days per index month: min / median / max	16 / 30 / 33
Index months failing the coverage rule	0
Share of headlines with a page title (%)	53.4
First index month with titles on $\geq$95% of headlines	2019-10
Distinct source domains	9,553
Share of the ten largest domains (%)	12.0

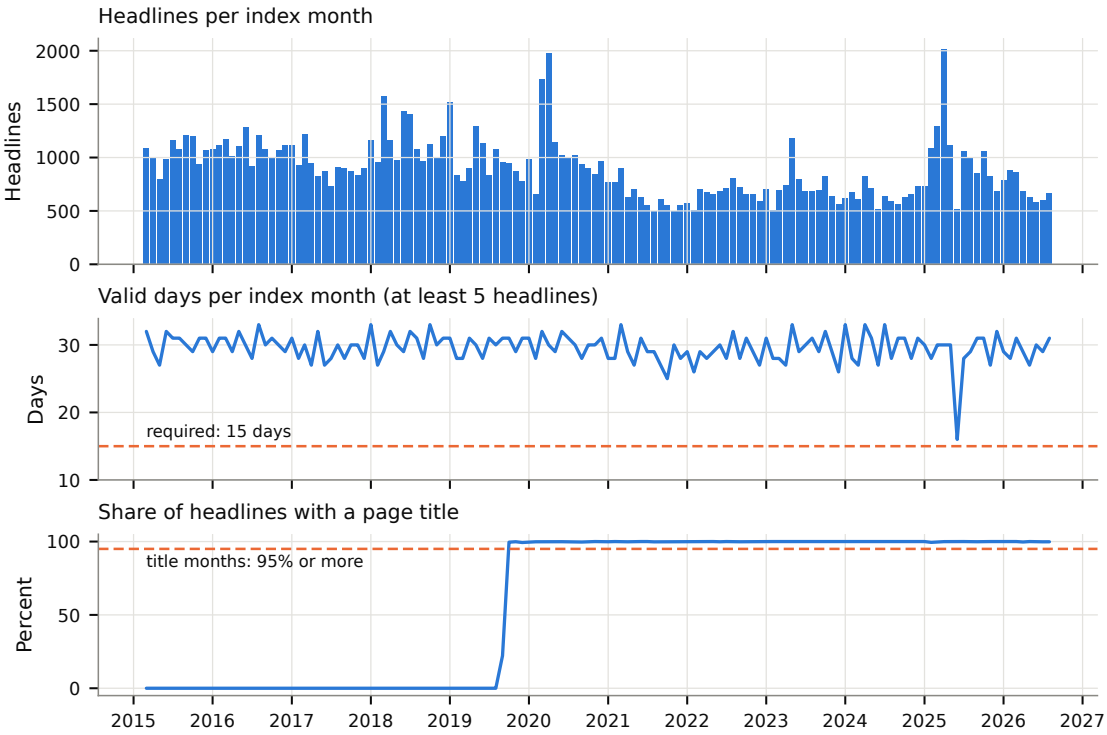


Figure 1: Coverage of the headline corpus by index month. Top: headlines that enter the index. Middle: days with enough headlines to enter the index, and the number of such days that an index month needs (dashed). Bottom: share of headlines with a page title, and the share from which a month counts as a title month in Section 6 (dashed).

Table 1 shows the result. Of 15,462,640 article rows read, 125,105 headlines remain, from 9,553 web domains; the ten largest domains supply 12.0% of them. An index month holds between 498 and 2,015 headlines, with a median of 870.5. Figure 1 shows the coverage month by month: the number of headlines, the number of days that enter the index, which falls to 16 in June 2025, and the share of headlines with a page title.

3.3 Published indices and scorer inputs

Three published series serve as benchmarks: the Daily News Sentiment Index of the Federal Reserve Bank of San Francisco [3], the news-based Economic Policy Uncertainty index for the United States [13] and, as a predictor of the S&P 500 only, the level of the consumer sentiment index. All three reach back to 1980 or earlier, which gives a long sample of 559 forecasts. After an initial window of 120 months it leaves 439 test months, from March 1990 to September 2026. The scorers use the word lists of Loughran and McDonald [15], the Financial PhraseBank [17] and the published FinBERT model [18].

4 Method

4.1 Sentiment scorers

A scorer maps one headline to a number that is higher for more positive text. We use four scorers that differ in kind. The first is a lexicon, a fixed list of words tagged positive or negative. With the finance word lists of Loughran and McDonald [15] the score is the number of positive words minus the number

of negative words, divided by their sum, and zero if the headline contains no listed word. Negation is not handled. The second is VADER, a general-purpose lexicon with rules for emphasis and negation that returns one number between -1 and 1 [16]. The third is FinBERT, a neural language model further trained on financial text and on labelled financial sentences [18]. Its score is the probability it assigns to the class positive minus the probability of the class negative. The fourth is a classifier that we train ourselves: a logistic regression on weighted counts of words and word pairs, fitted once to the 4,846 labelled sentences of the Financial PhraseBank [17] and scored like FinBERT. In five-fold cross-validation it classifies 76.2% of those sentences correctly, against 59.4% for always choosing the most frequent class. No scorer is tuned on the forecast targets.

Two further indices serve as comparisons. GDELT publishes its own tone score for the whole article, and we treat the index built from it as an external benchmark. Two large language models are added as a secondary comparison in Section 6.

4.2 Index construction

The daily index of a scorer is the mean score of the day's headlines, and a day with fewer than five headlines is missing. The monthly index is the mean of the daily index over a window that starts on the previous forecast date and ends the day before the current one. It needs at least 15 valid days, and all 138 index months meet this requirement. The primary predictor, fixed in advance, is the composite: each of the four monthly indices is standardised with its mean and standard deviation over the first 48 months only, and the four are averaged with equal weights. The composite was fixed before any experiment was run, so that no scorer could be chosen after the results were known. Its four members are built in different ways and need not agree. Each of them is therefore also reported on its own in every table, with a Bonferroni adjustment, and Section 5 reports how far they agree.

4.3 Forecast models and baselines

For the two continuous targets we compare four linear regressions, estimated by ordinary least squares. Let y_{t+1} be the target in month $t+1$ and x_t the sentiment index available on the forecast date at the end of month t. M0 forecasts with the mean of y so far, the historical mean. M1 regresses y_{t+1} on a constant and y_t, the own lag. M2 regresses y_{t+1} on a constant and x_t. M3 uses a constant, y_t and x_t. M0 and M1 are the baselines. The first asks whether the index beats a forecast that uses no predictor, the second whether it adds to the target's own past. For the direction we use logistic regressions with the same predictors, L0 to L3, where L0 predicts with the share of rising months so far and L1 uses the lagged excess return. A rise is predicted when the fitted probability is at least 0.5. The published indices enter the same models in place of x_t.

4.4 Evaluation

All models are estimated again every month on an expanding window. At each forecast date the model is fitted to all months up to that date and forecasts the next month, which is then added to the data. The forecast month is never part of the fitting data, so accuracy is measured only on months the model has not seen. This is what out-of-sample means here. The fit of a regression on its own estimation sample does not show it.

The main statistic is the out-of-sample R-squared [7],

$$R^2_{\mathrm{OS}} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y}_t)^2}, \tag{2}$$

where $\hat{y}_t$ is the forecast of the model, $\bar{y}_t$ that of the benchmark, and the sums run over the test months. It is the share by which the summed squared errors of the model are smaller than those of the benchmark. It is positive when the model is more accurate, zero when the two are equal and negative when the model is worse, with no lower bound.

A positive R^2_{OS} can arise by chance. M2 contains M0 as the special case in which the coefficient on the index is zero, and M3 contains M1 in the same way; such models are called nested. If that coefficient is truly zero, estimating it only adds noise, so the larger model is expected to have slightly larger errors in the test months. The test of Clark and West [12] removes this expected penalty and then asks whether the adjusted difference in squared errors is above zero. It is one-sided, and we use the 5% level. Because the test asks whether the index has predictive content, while R^2_{OS} measures realised accuracy, the test can reject while R^2_{OS} is at or below zero.

For the direction we report three quantities. The hit rate is the share of months in which the direction is called correctly. The test of Pesaran and Timmermann [22] asks whether the calls match the outcomes more often than chance; it is undefined, and shown as n/a, for a model that predicts the same direction every month. The Brier score is the mean squared error of the predicted probability, and we compare it with that of L0 by the test of Diebold and Mariano [23]. Intervals are 90% intervals from a stationary bootstrap [24], which resamples the test months 10,000 times in blocks with a mean length of six months, or twelve at the three-month horizon. At that horizon, where forecasts overlap, variances use the estimator of Newey and West [25].

4.5 Decision rule

The rule that maps results to an outcome was fixed before the experiments. For a continuous target the outcome is positive if the composite beats both baselines: M2 against M0 and M3 against M1 must each have R^2_{OS} above zero and a Clark-West p-value below 0.05. The outcome is mixed if exactly one comparison meets both conditions, and null otherwise. For the direction the outcome is positive if the hit rate of L2 is above that of L0 and the direction test has a p-value below 0.05, and null otherwise. The four individual scorers are secondary. Their p-values are multiplied by four, a Bonferroni adjustment for testing four scorers, and a null for the composite is reported as a null even if one scorer is significant. The p-values of all other comparisons in the paper are not adjusted.

4.6 Leakage controls

Leakage is information dated after a forecast was made that enters the forecast. Three rules prevent it. A headline is used only if the time at which GDELT recorded it is before the forecast date. A statistic about a month is published after that month, so every monthly series has a release lag. The policy uncertainty index is lagged one month, and so is consumer sentiment when it predicts the S&P 500. As its own lag, consumer sentiment enters with the latest survey value, whose final release can fall a few days after the forecast date; the text is still cut off at the forecast date, and Section 6 reports a variant without that value. The daily San Francisco Fed index is cut off seven days before the forecast date. The

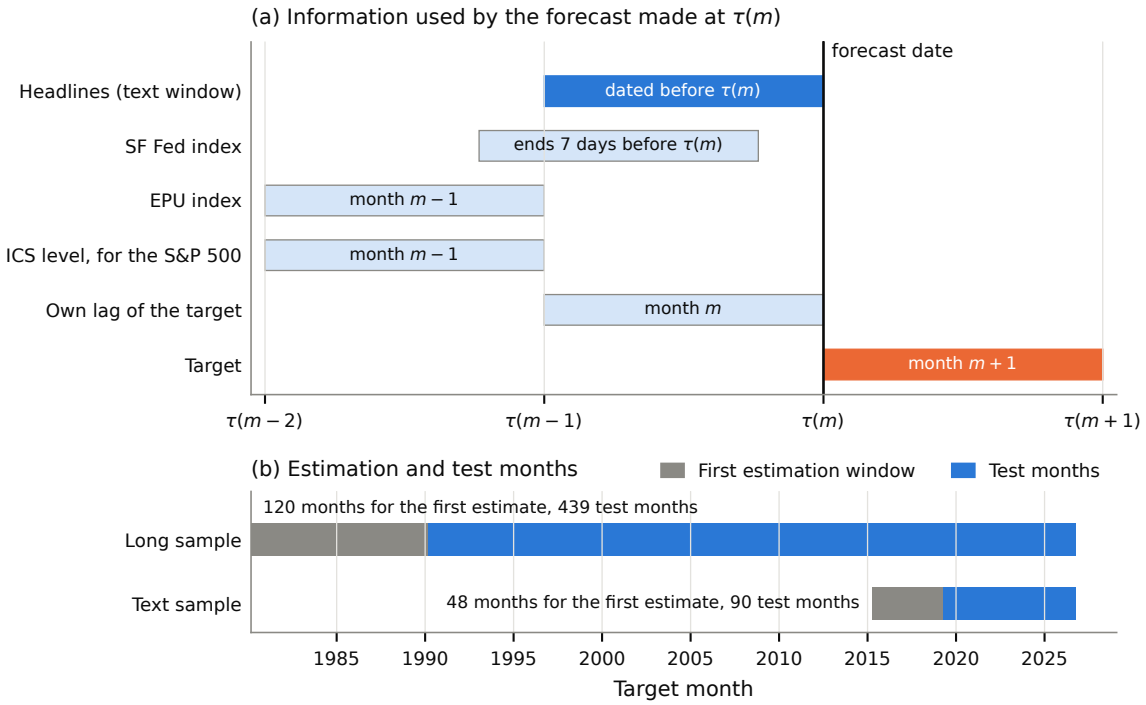


Figure 2: Timing. (a) The inputs of the forecast made on the forecast date $\tau(m)$, the last trading day of month m, and its target. Months are drawn with equal length. For consumer sentiment the own lag is the latest survey value, whose final release can fall a few days after $\tau(m)$. (b) The months used only for the first estimate and the test months, for the text sample and for the long sample of the published indices.

composite is standardised with the first 48 months only. Tests in the code fail if any headline is dated on or after its forecast date or if a series has no release lag. Figure 2 shows, for one forecast, the period that each input covers, and how the two samples divide into months for the first estimate and test months.

4.7 Changes to the plan

The plan was changed in five ways after it was fixed. First, before any request was sent to a language model, that comparison was extended from 200 headlines per month and one model to every headline and two models, with a split of the results at each model's training cutoff. Second, a check of a first complete run found that the downloader stopped looking for a replacement file after three attempts, a limit the plan does not contain, and that one robustness variant, the stricter timing for consumer sentiment in Section 6, was incomplete. Both were corrected, nine months of files were downloaded again, and all experiments were run again. Third, three tables were added or split off. Fourth, settings that the plan left open were fixed in the code, for example that a comparison needs at least 12 forecasts. Fifth, after the results were known, a simulation of the power of the main test (Section 6.2) and four figures drawn from the existing results were added. Before the first complete run, a trial run on three years of the corpus had produced the results for the published indices; no design choice was changed after it. Every outcome under the decision rule is the same in the first complete run and in the run reported here.

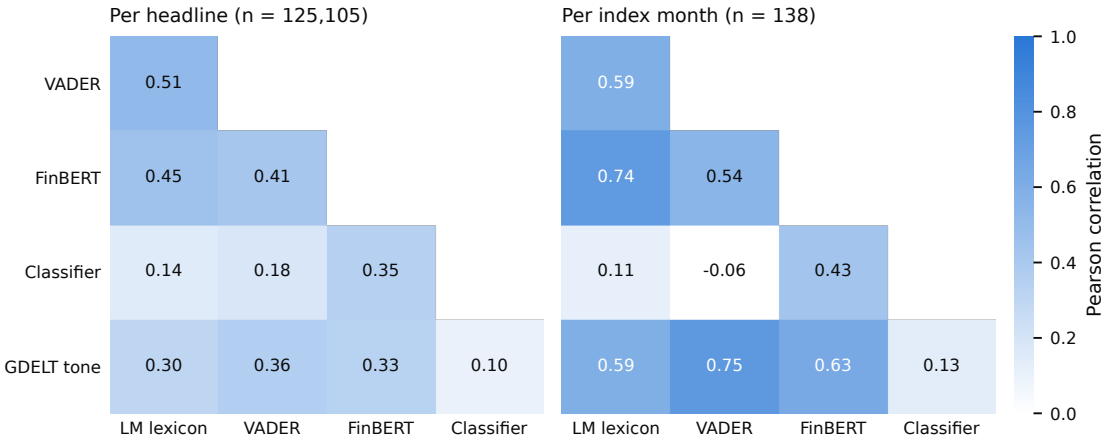


Figure 3: Agreement between scorers: Pearson correlation of the scores of single headlines (left) and of the monthly indices (right). The colour scale starts at zero.

5 Results

5.1 Measurement

Before the forecasts we ask whether the index is a sound measurement (RQ4). Table 2 and Figure 4 summarise the checks.

The four scorers agree only moderately on single headlines. The correlation between two scorers ranges from 0.14 for the finance lexicon and the classifier to 0.51 for the two lexicons. Averaging to months raises the agreement for some pairs and not for others: the monthly indices of the finance lexicon and of FinBERT correlate at 0.74, while those of VADER and of the classifier are almost unrelated (-0.06). Figure 3 shows every pair, with the index built from GDELT's tone as a fifth row.

The headline rebuilt from the web address is a close stand-in for the page title. On the 66,805 headlines that have a title, the score of the rebuilt headline and the score of the title correlate between 0.86 and 0.92, depending on the scorer. The monthly indices built from the two texts correlate at 0.96 or more over 83 months.

Two files a day are a small sample of each day's news, so part of the index is sampling noise. To measure it we build the index twice, once from each of the day's two files, and correlate the two halves; the Spearman-Brown formula then gives the reliability of the full index. It is 0.86 for the finance lexicon, 0.87 for VADER, 0.79 for FinBERT and 0.63 for the classifier.

These checks bear on the composite. The classifier has the lowest reliability of the four members, and its monthly index is almost unrelated to that of VADER, so the equal-weight mean combines indices that share only part of their variation. We keep the composite as the primary predictor because it was fixed in advance, and changing it now would be a choice made with the results in view. The tables report every member next to it, and no member on its own meets the conditions of the decision rule for any target.

The composite correlates at 0.48 with the San Francisco Fed index in levels and at 0.14 in monthly changes (Figure 4). Its correlation with the S&P 500 excess return is 0.19 for the same month and -0.15 for the next month. On the labelled sentences (Table 3) the two lexicons classify 63.2% and 54.3% correctly. FinBERT reaches 88.9%, but it was trained on these sentences.

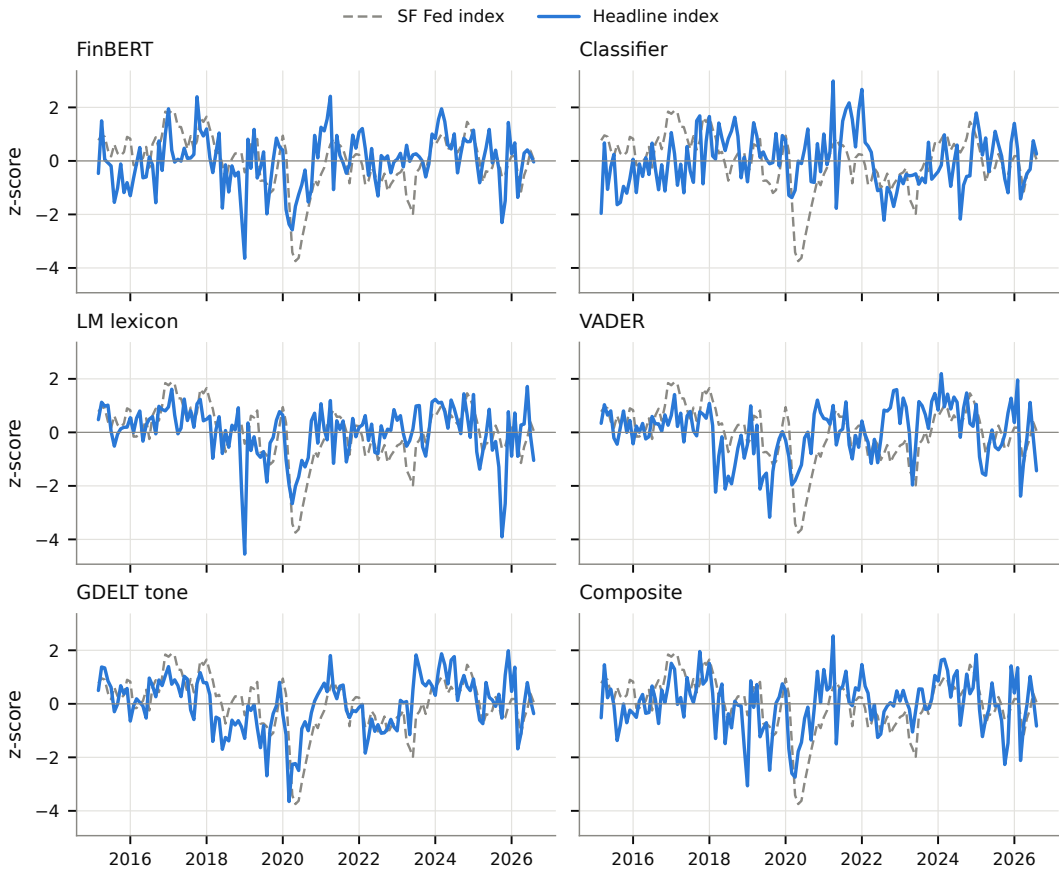


Figure 4: Monthly sentiment indices built from the headline corpus (solid) and the Daily News Sentiment Index of the Federal Reserve Bank of San Francisco (dashed), March 2015 to August 2026. Every series is standardised over the months shown.

Table 2: Measurement checks by index. Zero: share of headlines scored exactly zero. Split-half: reliability of the monthly index from the two daily files (Spearman-Brown). URL vs title: correlation between the scores of the rebuilt headline and of the page title, per headline and per month. Published series: correlation of the monthly index with the San Francisco Fed index and with the level of the Index of Consumer Sentiment (ICS). S&P 500 return: correlation with the excess return of the same and of the next month.

			URL vs title		Published series		S&P 500 return	
Index	Zero (%)	Split-half	Headline	Month	SF Fed	ICS	Same month	Next month
FinBERT	0.0	0.79	0.86	0.96	0.41	−0.02	0.20	−0.09
Classifier	0.0	0.63	0.87	0.97	0.13	0.14	0.11	−0.07
LM lexicon	62.6	0.86	0.92	0.98	0.49	0.15	0.07	−0.15
VADER	38.2	0.87	0.90	0.99	0.37	−0.11	0.17	−0.14
GDELT tone	4.1	0.92	n/a	n/a	0.60	0.01	0.18	−0.13
Composite	n/a	n/a	n/a	n/a	0.48	0.05	0.19	−0.15

Table 3: Accuracy and macro-F1 on the Financial PhraseBank with three classes. The classifier is evaluated by five-fold cross-validation. FinBERT was trained on these sentences. The scores of the two lexicons are turned into classes at -0.05 and 0.05.

Scorer	Accuracy (%)	Macro-F1 (%)
Classifier (5-fold)	76.2	66.8
LM lexicon	63.2	51.1
VADER	54.3	48.9
FinBERT (in sample)	88.9	88.2
Majority class	59.4	n/a

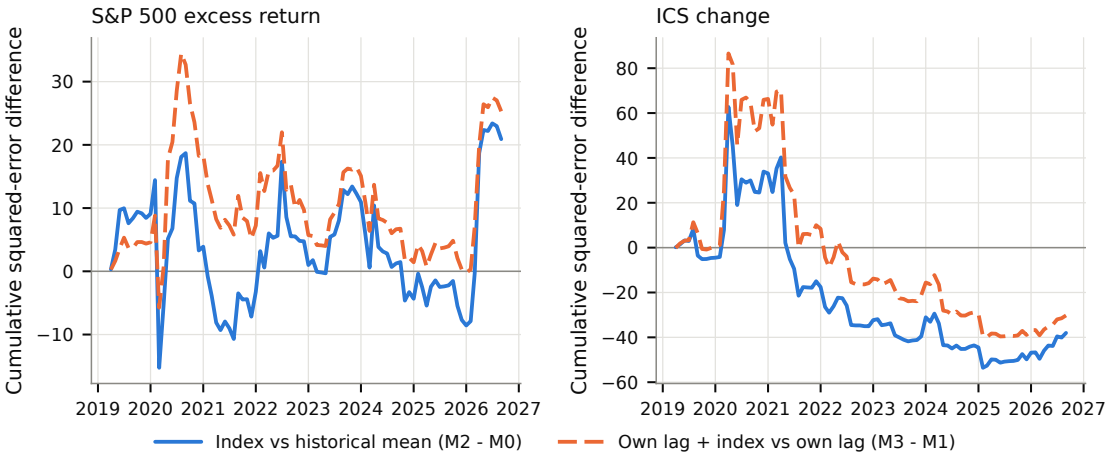


Figure 5: Cumulative difference in squared forecast errors over the test months, benchmark minus model with the composite index. A rising line means that the model with the index was more accurate than its benchmark in that month. Left: S&P 500 excess return. Right: change in the Index of Consumer Sentiment (ICS).

5.2 S&P 500 return and direction

Table 4 reports the forecasts of the S&P 500 excess return (RQ1). The own lag does not help: M1 has a larger squared error than the historical mean, with an R^2_{OS} of -2.61%. Adding the composite to the historical mean gives an R^2_{OS} of 1.03%, with a 90% interval from -2.44% to 4.80% and a Clark-West p-value of 0.149. Adding it to the own-lag model gives 1.21% (interval -1.97% to 4.41%, $p = 0.115$). Neither comparison meets the conditions of the rule, so the outcome is null.

No individual scorer changes this. Their R^2_{OS} against the historical mean lies between -1.31% for the classifier and 0.92% for VADER, and the smallest p-value before the Bonferroni adjustment is 0.134. The index built from GDELT's own tone gives -0.68%. Figure 5 shows when the composite gains and loses against each benchmark. The gain is not steady. Against the historical mean the largest loss in a single month falls in March 2020 and the largest gain in April 2026, and that one month supplies 19.0 of the total gain of 20.9 in squared percentage points. Figure 6 collects the point estimates and the intervals.

Table 5 reports the direction forecasts. Every model that uses a text index alone predicts a rise in each of the 90 test months. Its hit rate therefore equals the base rate of 63.3%, and the direction test is undefined. Figure 7 shows the fitted probabilities behind these calls: for the composite and for every individual index they stay at or above the threshold of 0.5 in all test months. The predicted probabilities are no more accurate than the base rate: the Brier score of the composite is 0.2377, against 0.2369 for L0.

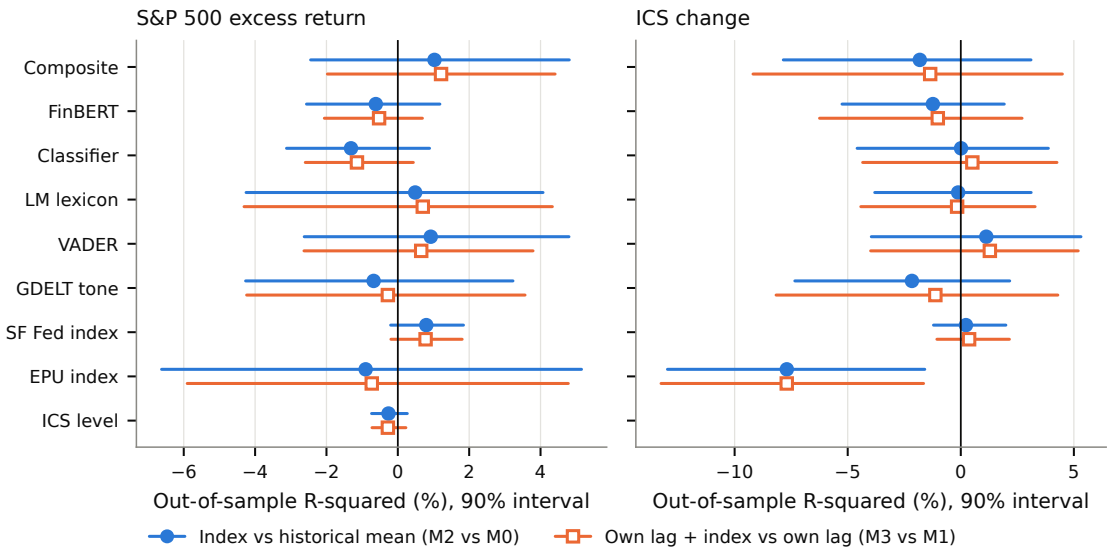


Figure 6: Out-of-sample R-squared with 90% bootstrap intervals, by predictor, on the 90 test months of the text sample. The models with published indices are estimated on data from 1980.

Table 4: One-month-ahead forecasts of the S&P 500 excess return. R^2_{OS} is the out-of-sample R-squared in percent, for a model with the predictor against the historical mean (M0) and for a model with the own lag and the predictor against the own-lag model (M1). Intervals are 90% bootstrap intervals. p is the one-sided Clark-West p-value and p adj. its Bonferroni adjustment for the four individual scorers. Upper block: text indices, 90 test months. Lower blocks: published indices.

Predictor	R^2_{OS} vs M0 (%)	90% interval	p	p adj.	R^2_{OS} vs M1 (%)	90% interval	p	p adj.
Composite	1.03	[−2.44, 4.80]	0.149		1.21	[−1.97, 4.41]	0.115	
FinBERT	−0.62	[−2.56, 1.18]	0.542	1.000	−0.53	[−2.06, 0.69]	0.627	1.000
Classifier	−1.31	[−3.12, 0.89]	0.737	1.000	−1.15	[−2.59, 0.43]	0.817	1.000
LM lexicon	0.49	[−4.25, 4.07]	0.134	0.534	0.70	[−4.31, 4.33]	0.129	0.518
VADER	0.92	[−2.62, 4.80]	0.170	0.681	0.66	[−2.63, 3.79]	0.217	0.866
GDELT tone	−0.68	[−4.27, 3.23]	0.392		−0.28	[−4.24, 3.57]	0.364	
Published indices and their benchmarks, estimated from 1980: same 90 test months								
SF Fed index	0.80	[−0.20, 1.84]	0.076		0.78	[−0.19, 1.80]	0.078	
EPU index	−0.90	[−6.62, 5.15]	0.040		−0.73	[−5.90, 4.78]	0.042	
ICS level	−0.26	[−0.74, 0.27]	0.780		−0.28	[−0.72, 0.23]	0.806	
Published indices and their benchmarks, estimated from 1980: long sample, 439 test months								
SF Fed index	−0.79	[−1.98, 0.36]	0.516		−0.72	[−1.86, 0.33]	0.496	
EPU index	−0.34	[−3.23, 2.67]	0.002		−0.27	[−3.04, 2.55]	0.002	
ICS level	−0.55	[−1.10, −0.03]	0.915		−0.54	[−1.04, −0.07]	0.931	

Table 5: One-month-ahead forecasts of the direction of the S&P 500 from logistic models. Hit rate: share of test months with the direction called correctly. Predicted up: share of months in which a rise is predicted. PT p: Pesaran-Timmermann test, n/a when the model predicts the same direction every month. Brier: mean squared error of the predicted probability. DM p: Diebold-Mariano test of the Brier score against the L0 of the same block.

Predictor	Hit rate (%)	90% interval	Predicted up (%)	PT p	PT p adj.	Brier	DM p vs L0
Base rate (L0)	63.3	[56.7, 70.0]	100.0	n/a		0.2369	n/a
Lagged return (L1)	62.2	[55.6, 68.9]	98.9	0.779		0.2420	0.134
Composite	63.3	[56.7, 70.0]	100.0	n/a		0.2377	0.818
FinBERT	63.3	[56.7, 70.0]	100.0	n/a	n/a	0.2417	0.048
Classifier	63.3	[56.7, 70.0]	100.0	n/a	n/a	0.2401	0.094
LM lexicon	63.3	[56.7, 70.0]	100.0	n/a	n/a	0.2407	0.530
VADER	63.3	[56.7, 70.0]	100.0	n/a	n/a	0.2372	0.967
GDELT tone	63.3	[56.7, 70.0]	100.0	n/a		0.2392	0.663
Published indices and their benchmarks, estimated from 1980: same 90 test months							
Base rate (L0)	63.3	[56.7, 70.0]	100.0	n/a		0.2327	n/a
Lagged return (L1)	63.3	[56.7, 70.0]	100.0	n/a		0.2340	0.465
SF Fed index	63.3	[56.7, 70.0]	100.0	n/a		0.2321	0.606
EPU index	63.3	[56.7, 70.0]	100.0	n/a		0.2392	0.533
ICS level	63.3	[56.7, 70.0]	100.0	n/a		0.2339	0.049
Published indices and their benchmarks, estimated from 1980: long sample, 439 test months							
Base rate (L0)	63.8	[59.7, 67.9]	100.0	n/a		0.2323	n/a
Lagged return (L1)	63.8	[59.7, 67.7]	100.0	n/a		0.2329	0.363
SF Fed index	62.6	[58.8, 66.7]	96.1	0.532		0.2353	0.126
EPU index	63.1	[59.0, 67.0]	98.9	0.776		0.2340	0.622
ICS level	63.8	[59.7, 67.7]	100.0	n/a		0.2333	0.139

The outcome is null.

5.3 Consumer sentiment

Table 6 reports the forecasts of the change in consumer sentiment (RQ2). The own lag is again worse than the historical mean, with an R^2_{OS} of -6.37%. The composite improves on neither baseline. Its R^2_{OS} is -1.81% against the historical mean ($p = 0.450$) and -1.35% when it is added to the own lag ($p = 0.387$). The outcome is null. Among the individual scorers VADER has the largest R^2_{OS}, 1.13%, with a p-value of 0.120 before the adjustment and 0.481 after it.

5.4 Published indices

The lower blocks of Tables 4 to 6 repeat the exercise with the published indices (RQ3). These models and their benchmarks are estimated on data from 1980, so their baselines differ from those of the text models, which start in 2015. Over the long sample of 439 test months none of the three indices lowers the squared error for the S&P 500 return. The R^2_{OS} against the historical mean is -0.79% for the San Francisco Fed index, -0.34% for the policy uncertainty index and -0.55% for the level of consumer sentiment. For the policy uncertainty index the Clark-West test rejects ($p = 0.002$) although its R^2_{OS} is negative, the case described in Section 4. For consumer sentiment the San Francisco Fed index gives -0.55% against the historical mean and -0.58% when it is added to the own lag. On the 90 months of the text sample the San Francisco Fed index gives 0.80% for the S&P 500 return ($p = 0.076$) and 0.23% for consumer sentiment ($p = 0.337$).

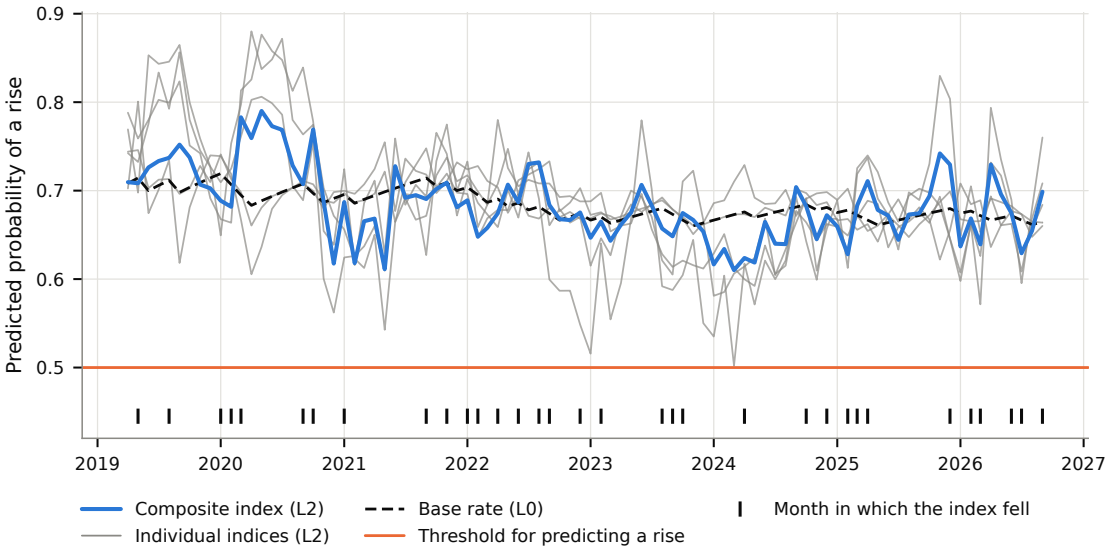


Figure 7: Predicted probability that the S&P 500 rises in the next month, from the logistic models with one text index (L2) and from the base-rate model (L0), over the 90 test months. A rise is predicted when the probability is at least 0.5. The marks at the bottom show the months in which the index fell.

Table 6: One-month-ahead forecasts of the change in the Index of Consumer Sentiment. Columns and blocks as in Table 4.

Predictor	R^2_{OS} vs M0 (%)	90% interval	p	p adj.	R^2_{OS} vs M1 (%)	90% interval	p	p adj.
Composite	−1.81	[−7.85, 3.10]	0.450		−1.35	[−9.19, 4.49]	0.387	
FinBERT	−1.23	[−5.25, 1.92]	0.483	1.000	−1.02	[−6.24, 2.71]	0.441	1.000
Classifier	0.01	[−4.58, 3.87]	0.331	1.000	0.51	[−4.34, 4.25]	0.239	0.956
LM lexicon	−0.12	[−3.80, 3.11]	0.287	1.000	−0.16	[−4.42, 3.29]	0.318	1.000
VADER	1.13	[−3.96, 5.31]	0.120	0.481	1.29	[−3.98, 5.19]	0.108	0.431
GDELT tone	−2.16	[−7.34, 2.16]	0.469		−1.12	[−8.17, 4.29]	0.368	
Published indices and their benchmarks, estimated from 1980: same 90 test months								
SF Fed index	0.23	[−1.21, 1.99]	0.337		0.37	[−1.06, 2.15]	0.300	
EPU index	−7.69	[−12.97, −1.59]	0.693		−7.70	[−13.25, −1.65]	0.693	
Published indices and their benchmarks, estimated from 1980: long sample, 439 test months								
SF Fed index	−0.55	[−2.61, 1.23]	0.262		−0.58	[−2.61, 1.19]	0.269	
EPU index	−1.11	[−4.13, 1.79]	0.086		−1.11	[−4.20, 1.82]	0.086	

Table 7: The composite index added to models that already contain other predictors. R^2_{OS} compares the model with the composite against the same model without it, on 90 test months.

Target	Benchmark holds	R^2_{OS} vs benchmark (%)	90% interval	p
S&P 500 excess return	SF Fed index	−0.79	[−3.25, 1.86]	0.594
S&P 500 excess return	SF Fed index, EPU index, ICS level	−0.94	[−2.74, 1.34]	0.660
ICS change	own lag, SF Fed index	0.46	[−9.00, 7.39]	0.152
ICS change	own lag, SF Fed index, EPU index	0.30	[−8.92, 6.65]	0.178
ICS change	own lag, lagged S&P 500 return	−1.27	[−4.24, 1.28]	0.617

Table 8: Robustness of the composite index. n: number of test months. R^2_{OS} and hit rates are in percent. Hit rate / base: hit rate of the logistic model with the composite and of the base-rate model. Title months: index months in which at least 95% of the headlines have a page title. ICS: Index of Consumer Sentiment.

		S&P 500			ICS change			
Variant	n	R^2_{OS} vs M0	p	Hit rate / base	R^2_{OS} vs M0	p	R^2_{OS} vs M1	p
Initial window 36	102	0.75	0.142	64.7 / 65.7	−1.35	0.354	−0.43	0.264
Initial window 60	78	2.14	0.062	64.1 / 64.1	−2.66	0.528	−2.94	0.536
Rolling window 48	90	0.97	0.153	63.3 / 63.3	−2.34	0.371	−1.33	0.301
Index in first differences	89	−1.72	0.889	61.8 / 62.9	1.63	0.091	1.32	0.092
Headline-weighted index	90	−0.06	0.253	63.3 / 63.3	−1.53	0.362	−1.24	0.324
Page-title index (title months)	47	2.14	0.175	63.8 / 63.8	0.53	0.284	−1.17	0.533
URL-headline index (title months)	47	1.59	0.191	63.8 / 63.8	−0.18	0.343	−1.73	0.619
Raw log return	90	0.92	0.160					
Simple excess return	90	1.33	0.117					
ICS own lag, release lag 1	88				−2.80	0.600	−1.90	0.373
Three-month horizon	86	5.69	0.049		0.85	0.181	−0.30	0.449

Table 7 asks whether the composite adds to models that already contain the published indices. Added to a model with the San Francisco Fed index it gives an R^2_{OS} of −0.79% for the S&P 500 return. For consumer sentiment, added to the own lag and the San Francisco Fed index, it gives 0.46% ($p = 0.152$), and added to the own lag and the lagged S&P 500 return it gives −1.27%.

6 Robustness and limitations

6.1 Robustness

Table 8 repeats the three main experiments under other choices. With an initial window of 36 or 60 months, or a rolling window of 48 months, the R^2_{OS} of the composite for the S&P 500 return is 0.75%, 2.14% and 0.97%, and no outcome changes. No outcome changes either when the index enters in first differences, when the monthly index weights headlines and not days, or when the index is built from page titles on the 47 test months that have them. With the raw log return or a simple excess return as the target, the R^2_{OS} is 0.92% and 1.33%. For consumer sentiment a stricter variant, in which the latest survey value is assumed unknown on the forecast date, gives −1.90% when the composite is added to the own lag.

At a three-month horizon the result for the S&P 500 differs. The composite gives an R^2_{OS} of 5.69% against the historical mean, with an interval from −8.70% to 16.02% and a p-value of 0.049, and 4.25% when it is added to the own lag ($p = 0.087$). Under the rule this outcome is mixed. The 86 test months

Table 9: The sentiment index of a large language model in the forecasting models, by model. The second and third row of each block split the test months by whether the whole text window lies after the month in which the training data of the model end, as stated by its provider. R^2_{OS} and hit rates are in percent. n/a: fewer than 12 forecasts.

		S&P 500			ICS change			
Model and test months	n	R^2_{OS} vs M0	p	Hit rate / base	R^2_{OS} vs M0	p	R^2_{OS} vs M1	p
claude-haiku-4-5, all test months	90	−1.51	0.459	62.2 / 63.3	0.37	0.185	0.71	0.183
after the cutoff month (2025-07)	12	4.31	0.216	50.0 / 50.0	13.38	0.029	11.41	0.019
other test months	78	−1.98	0.511	64.1 / 65.4	−0.64	0.301	−0.05	0.262
claude-sonnet-5-5, all test months	90	−1.86	0.637	62.2 / 63.3	0.29	0.193	0.72	0.184
after the cutoff month (2026-06)	1	n/a	n/a	n/a	n/a	n/a	n/a	n/a
other test months	89	−1.80	0.628	62.9 / 64.0	0.09	0.213	0.58	0.196

of this experiment overlap, and the plan lists the three-month horizon as a secondary analysis. For consumer sentiment the outcome at three months is null.

6.2 Power of the main test

The analysis in this subsection was not part of the plan. It was designed after the outcomes were known and has no outcome under the decision rule. It asks how large the predictive content of the composite would have to be for the main test to find it on 90 test months. We hold the composite at its observed values and simulate the target for a range of true R-squared values, where the true R-squared is the share of the variance of the target that the index explains in the simulated data. The noise is drawn with replacement from the standardised observed target, which keeps its tails. For each value, 10,000 simulated samples go through the same expanding-window forecasts of M0 and M2 and the same Clark-West test as the data. Power is the share of samples in which M2 meets both conditions of the decision rule against M0: an R^2_{OS} above zero and a p-value below 0.05.

For the S&P 500 excess return the power is 14.3% at a true R-squared of 1%, 26.9% at 2% and 61.3% at 5%. It reaches 50% at a true R-squared of 4.0% and 80% at 8.0%. When the index has no predictive content, 3.2% of the samples meet both conditions. For the change in consumer sentiment the power also reaches 80% at 8.0%; the two simulations differ only in the distribution of the noise. The simulation covers the comparison with the historical mean at the one-month horizon. It assumes a linear relation that is constant over the sample and noise that is independent across months.

6.3 Language models as scorers

Table 9 reports a secondary comparison in which the scorer is a large language model. Two commercial models, `claude-haiku-4-5` and `claude-sonnet-5-5`, label each headline with an integer from -2 to 2, and the index is built as for the other scorers. This index is outside the composite and the decision rule, and its p-values are not adjusted. Over all test months the index of neither model improves on the baselines for the S&P 500 return, with an R^2_{OS} of -1.51% for the first model and -1.86% for the second. For consumer sentiment, added to the own lag, the two give 0.71% ($p = 0.183$) and 0.72% ($p = 0.184$).

A language model may have been trained on text written after the headlines it scores [20]. The training data of the first model end in July 2025 according to its provider, and 12 test months have a text window that lies wholly after that month. On these months its index gives 4.31% for the S&P 500

return ($p = 0.216$) and 11.41% for consumer sentiment added to the own lag, with an interval from 6.14% to 21.08% and a p-value of 0.019. Three facts qualify this number. The sample is the smallest that our code evaluates. The forecasting models are still estimated on index values from before the cutoff. The comparison is one of many in this paper. For the second model only 1 test month has a text window wholly after the cutoff, so the comparison cannot be made.

6.4 Limitations

The test covers 90 months, which include the fall and recovery of the market in March and April 2020. A sample of this size has low power. By the simulation of Section 6.2 the test reaches a power of 80% only from a true R-squared of 8.0%, which is far above the small values that can matter to an investor [7], and that simulation was itself designed after the results were known. Every headline number therefore comes with an interval. The study covers one country, one stock index and one survey series. The S&P 500 series is a price index without dividends, and the release dates of the consumer sentiment survey were not verified by machine.

The corpus has five limits. Headlines are rebuilt from web addresses and are not the titles that readers saw. Two files a day cover a small part of each day's news. The set of sources changes over the sample: 54.5% of the headlines of 2025 come from domains that also appear in 2015, and 3 of the 20 largest domains are the same in both years. The archive has no readable file for 17 days in June and July 2025, so the index month of June 2025 rests on 16 days. The list of economy and market terms was fixed by hand in 2026, after the sample period, and was not validated against human labels.

Several scorers and targets are tested. One primary predictor was fixed in advance and the four individual scorers carry a Bonferroni adjustment, but the p-values of the tone index, the published indices, the robustness checks and the language models are unadjusted. The published indices are used as downloaded in October 2026, and revisions of their history were not examined. For the language models, one request holds up to 40 headlines, so a label can be produced next to headlines dated after a forecast date, and the training cutoffs are statements of the provider that we did not check. That comparison is also the only part of the study that needs a paid service.

7 Conclusion

We tested whether sentiment measured from free news headlines improves one-month-ahead forecasts of the S&P 500 and of consumer sentiment, with the plan and the decision rule fixed in advance. Under that rule the outcome is null for the S&P 500 excess return, for its direction and for the change in consumer sentiment. The measurement checks describe an index that is reliable given the sampling, with a split-half reliability between 0.63 and 0.87, that agrees with an index built from page titles and that correlates with a published news sentiment index. Published indices do not lower the squared error of the S&P 500 forecasts over 439 test months either.

The null has a limited meaning. It agrees with the finding that most predictors of the aggregate stock return fail out of sample [5, 6]. It does not show that news sentiment carries no information. The 90% interval for the S&P 500 return includes zero and also values that would be large by the standards of that literature, in which a small out-of-sample R-squared can matter to an investor [7], and 90 test months cannot separate the two. The power simulation says the same from the other side: the test

finds a true R-squared of 1% in 14.3% of simulated samples and needs 8.0% for a power of 80%. The null speaks against a large effect and leaves a small one open. The measurement checks narrow the reading in one respect: sampling noise does not dominate the index, and its use of web addresses in place of titles changes little.

Two results fall outside the null. At a three-month horizon the outcome for the S&P 500 return is mixed, and the index of one language model lowers the squared error for consumer sentiment on the 12 months after its training cutoff. Both rest on small or overlapping samples and neither is the primary test, so we report them as candidates for a future test fixed in advance and not as findings. Within the same design the natural extensions are a larger share of each day's news, the text of the articles, and the daily horizon at which earlier studies report effects [2, 9]. The study reruns from free data with one command, so the test can be repeated as months are added to the sample.

Acknowledgements

This research received no specific funding.

Use of AI tools. During the preparation of this work, the author used Claude (Anthropic). The tool was used to suggest related literature during background research, to propose a rudimentary outline of the sections of the paper and to organise the code repository that accompanies the paper. It was not used to generate text for the paper: all text was written by the author. The author takes full responsibility for the publication's content. The two language models evaluated as sentiment scorers in Section 6 are objects of study and were not used for writing.

Statements

Data availability. The data come from the public sources named in Section 3. The code, the result files behind every table and figure, and the command that reruns the study from raw downloads are available at `https://github.com/vignesh-nagarajan-vn/SentCast`. The language model labels used in Section 6 are reproduced from cached files, because the model `claude-haiku-4-5` may no longer be served. These labels are not part of the repository and are available from the author on request.

Funding. This research received no specific funding.

Conflict of interest. The author declares no conflict of interest.

References

[1] M. Gentzkow, B. Kelly, M. Taddy, Text as Data, Journal of Economic Literature 57 (2019) 535–574. doi:10.1257/jel.20181020.

[2] P. C. Tetlock, Giving Content to Investor Sentiment: The Role of Media in the Stock Market, The Journal of Finance 62 (2007) 1139–1168. doi:10.1111/j.1540-6261.2007.01232.x.

[3] A. H. Shapiro, M. Sudhof, D. J. Wilson, Measuring news sentiment, Journal of Econometrics 228 (2022) 221–243. doi:10.1016/j.jeconom.2020.07.053.

[4] V. Nagarajan, A. Mittal, PEBSA: Predicting Economic Behavior via Sentiment Analysis, International Journal of Engineering and Computer Science 13 (2024) 26656–26676. doi:10.18535/ijecs/v13i12.4950.

[5] I. Welch, A. Goyal, A Comprehensive Look at The Empirical Performance of Equity Premium Prediction, Review of Financial Studies 21 (2008) 1455–1508. doi:10.1093/rfs/hhm014.

[6] A. Goyal, I. Welch, A. Zafirov, A Comprehensive 2022 Look at the Empirical Performance of Equity Premium Prediction, The Review of Financial Studies 37 (2024) 3490–3557. doi:10.1093/rfs/hhae044.

[7] J. Y. Campbell, S. B. Thompson, Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?, Review of Financial Studies 21 (2008) 1509–1531. doi:10.1093/rfs/hhm055.

[8] D. Huang, F. Jiang, J. Tu, G. Zhou, Investor Sentiment Aligned: A Powerful Predictor of Stock Returns, Review of Financial Studies 28 (2015) 791–837. doi:10.1093/rfs/hhu080.

[9] D. García, Sentiment during Recessions, The Journal of Finance 68 (2013) 1267–1300. doi:10.1111/jofi.12027.

[10] J. Bollen, H. Mao, X. Zeng, Twitter mood predicts the stock market, Journal of Computational Science 2 (2011) 1–8. doi:10.1016/j.jocs.2010.12.007.

[11] Z. T. Ke, B. Kelly, D. Xiu, Predicting Returns With Text Data, Working Paper 26186, National Bureau of Economic Research, 2019. doi:10.3386/w26186.

[12] T. E. Clark, K. D. West, Approximately normal tests for equal predictive accuracy in nested models, Journal of Econometrics 138 (2007) 291–311. doi:10.1016/j.jeconom.2006.05.023.

[13] S. R. Baker, N. Bloom, S. J. Davis, Measuring Economic Policy Uncertainty, The Quarterly Journal of Economics 131 (2016) 1593–1636. doi:10.1093/qje/qjw024.

[14] E. Kalamara, A. Turrell, C. Redl, G. Kapetanios, S. Kapadia, Making text count: Economic forecasting using newspaper text, Journal of Applied Econometrics 37 (2022) 896–919. doi:10.1002/jae.2907.

[15] T. Loughran, B. McDonald, When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks, The Journal of Finance 66 (2011) 35–65. doi:10.1111/j.1540-6261.2010.01625.x.

[16] C. Hutto, E. Gilbert, VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text, Proceedings of the International AAAI Conference on Web and Social Media 8 (2014) 216–225. doi:10.1609/icwsm.v8i1.14550.

[17] P. Malo, A. Sinha, P. Korhonen, J. Wallenius, P. Takala, Good debt or bad debt: Detecting semantic orientations in economic texts, Journal of the Association for Information Science and Technology 65 (2014) 782–796. doi:10.1002/asi.23062.

[18] D. Araci, FinBERT: Financial Sentiment Analysis with Pre-trained Language Models, arXiv:1908.10063, 2019. URL: https://arxiv.org/abs/1908.10063.

[19] A. Lopez-Lira, Y. Tang, Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models, arXiv:2304.07619, 2023. URL: https://arxiv.org/abs/2304.07619.

[20] P. Glasserman, C. Lin, Assessing Look-Ahead Bias in Stock Return Predictions Generated By GPT Sentiment Analysis, arXiv:2309.17322, 2023. URL: https://arxiv.org/abs/2309.17322.

[21] K. Leetaru, P. A. Schrodt, GDELT: Global Data on Events, Location and Tone, 1979–2012, Paper presented at the International Studies Association meetings, San Francisco, April 2013, 2013. URL: http://data.gdeltproject.org/documentation/ISA.2013.GDELT.pdf.

[22] M. H. Pesaran, A. Timmermann, A Simple Nonparametric Test of Predictive Performance, Journal of Business & Economic Statistics 10 (1992) 461–465. doi:10.1080/07350015.1992.10509922.

[23] F. X. Diebold, R. S. Mariano, Comparing Predictive Accuracy, Journal of Business & Economic Statistics 13 (1995) 253–263. doi:10.1080/07350015.1995.10524599.

[24] D. N. Politis, J. P. Romano, The Stationary Bootstrap, Journal of the American Statistical Association 89 (1994) 1303–1313. doi:10.1080/01621459.1994.10476870.

[25] W. K. Newey, K. D. West, A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix, Econometrica 55 (1987) 703–708. doi:10.2307/1913610.